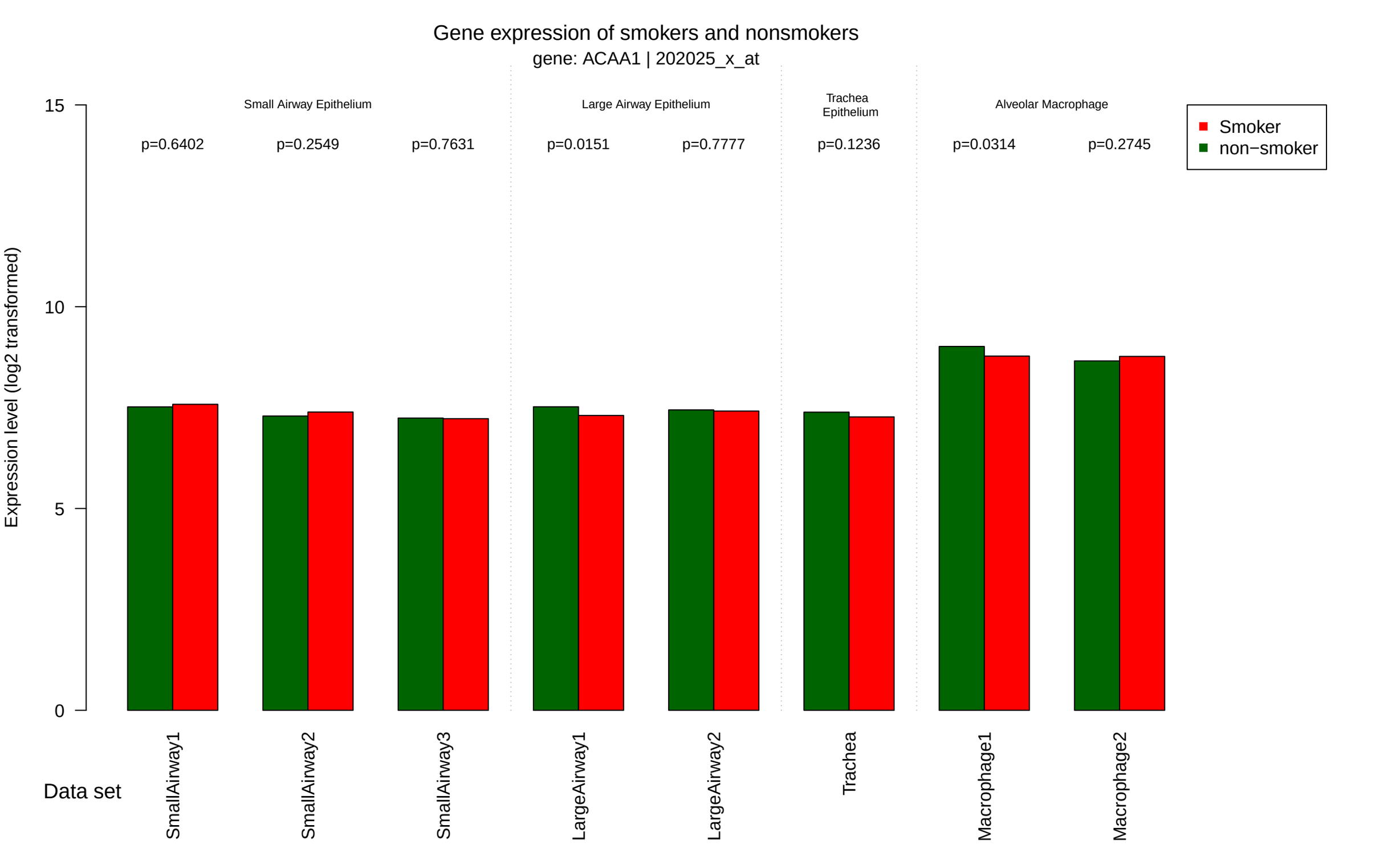
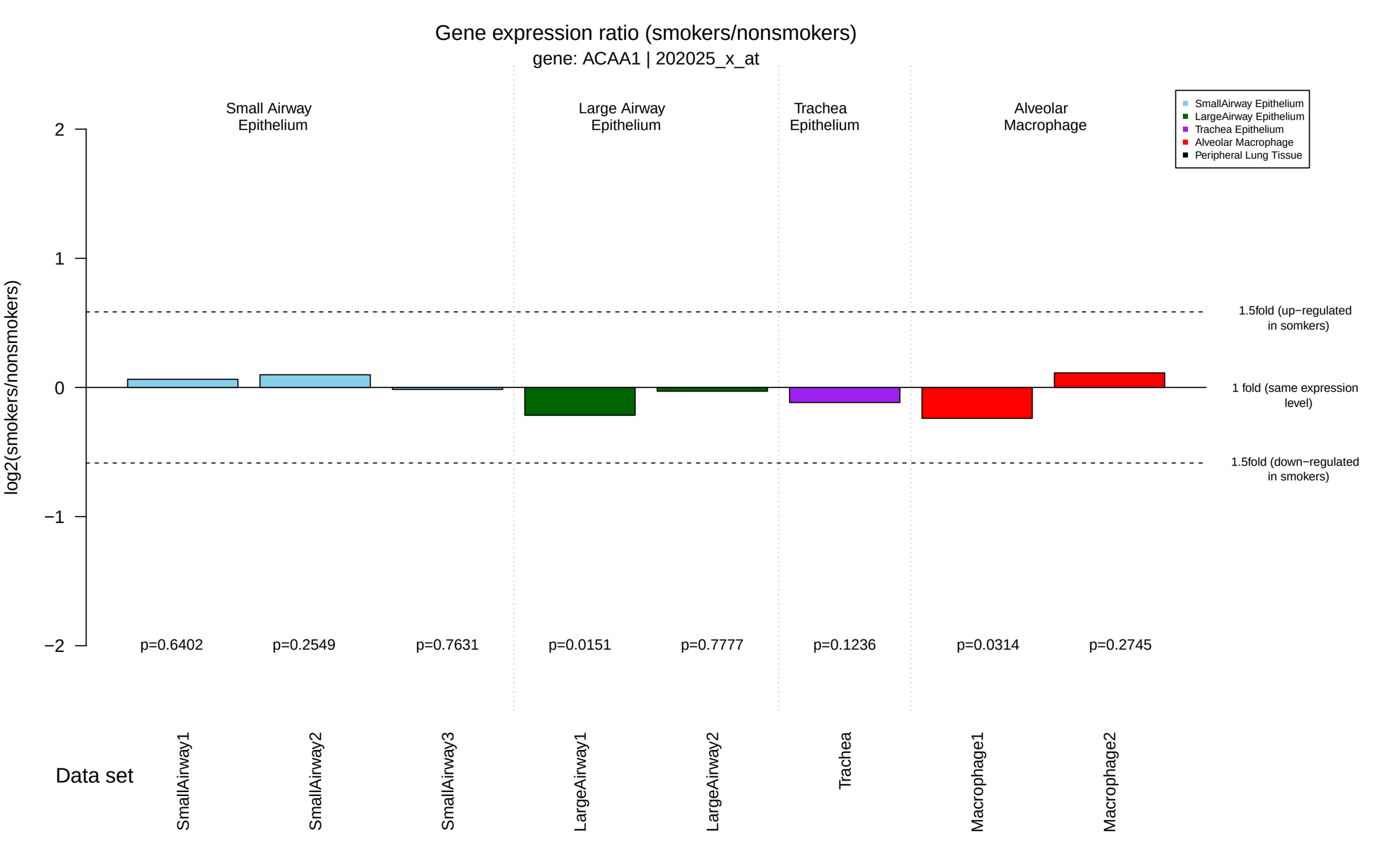
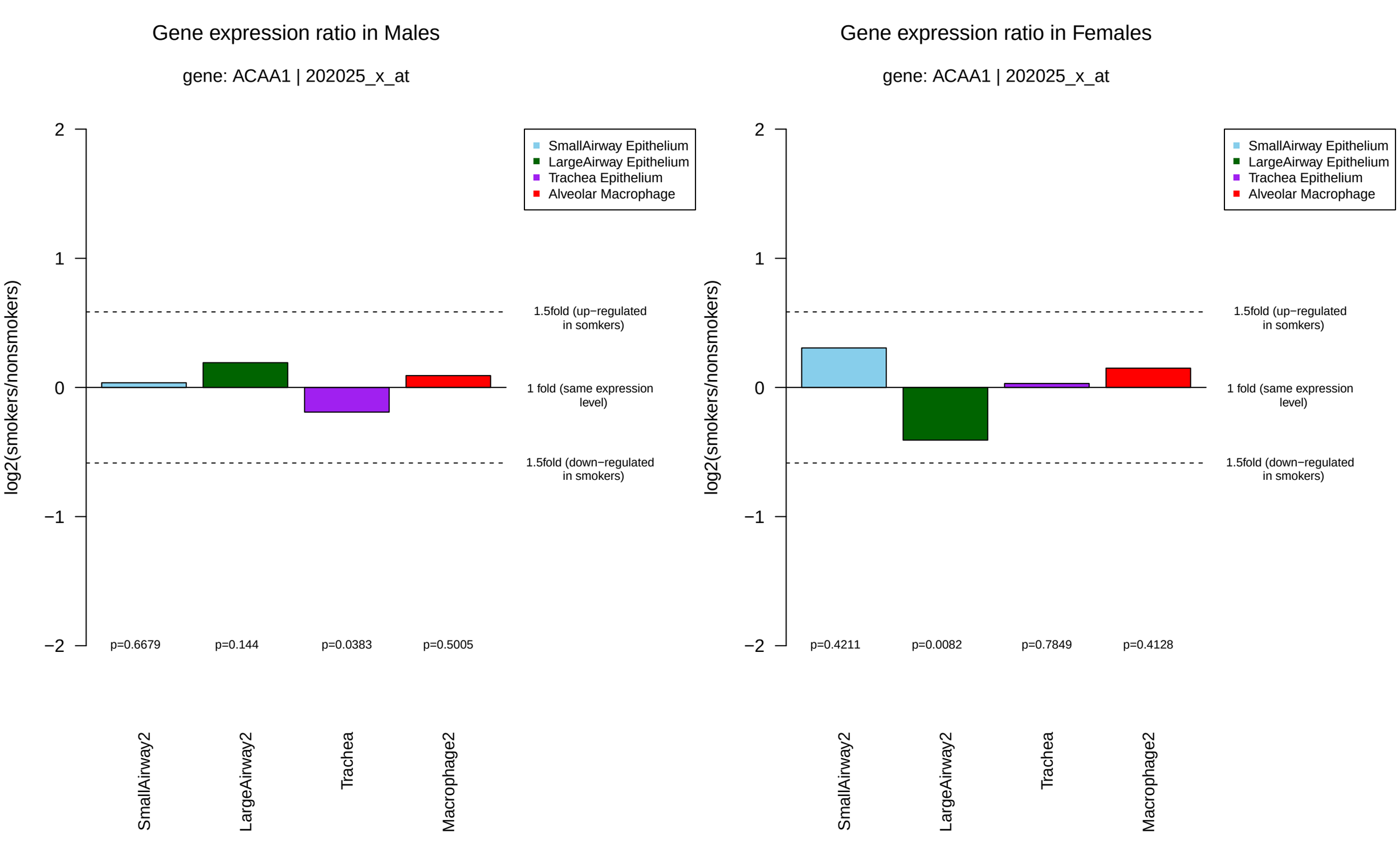
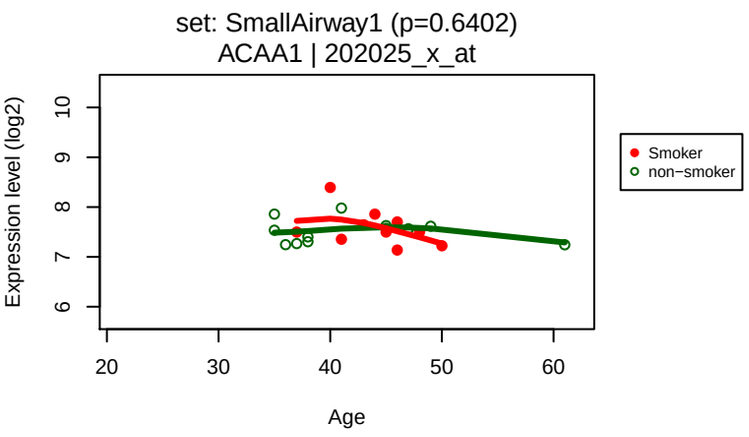
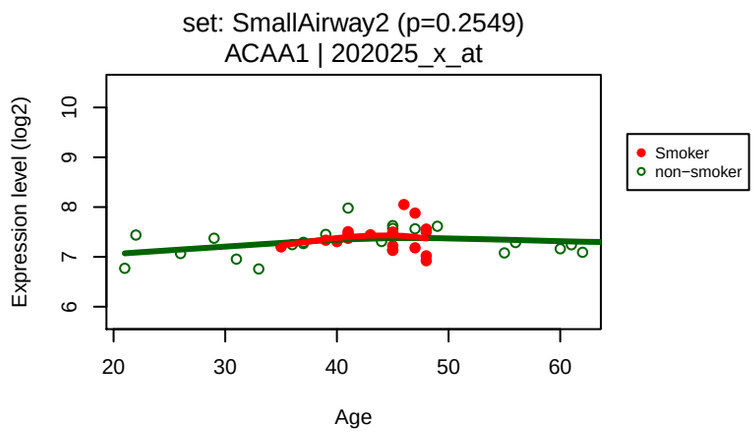
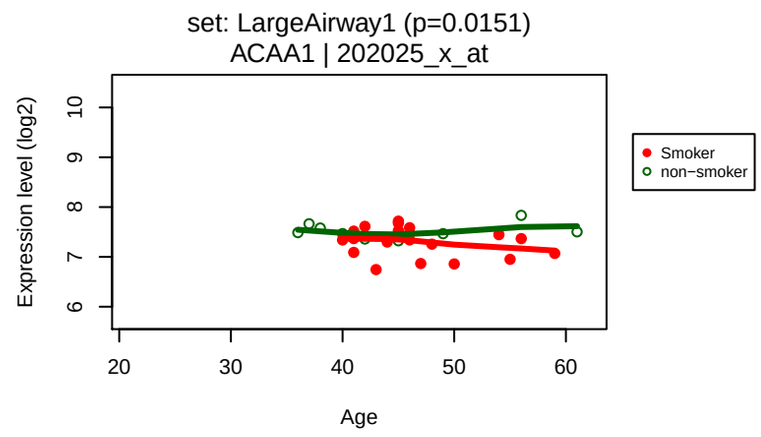

Gene expression of smokers and nonsmokers
gene: ACAA1 | 202025_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6402
p=0.2549
p=0.7631
p=0.0151
p=0.7777
p=0.1236
p=0.0314
p=0.2745
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

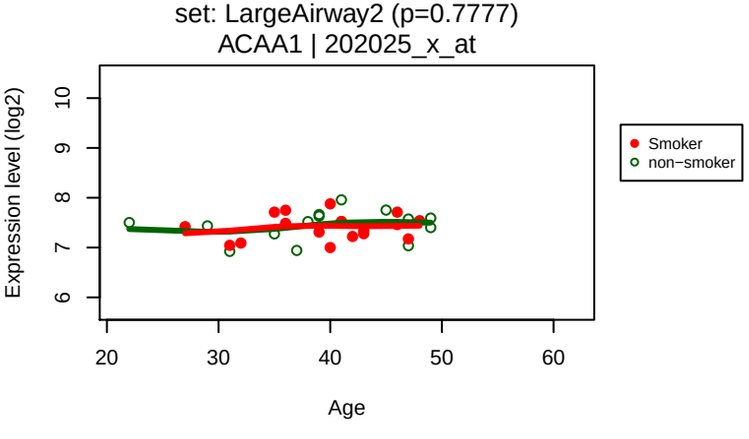

Gene expression ratio (smokers/nonsmokers)
gene: ACAA1 | 202025_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6402
p=0.2549
p=0.7631
p=0.0151
p=0.7777
p=0.1236
p=0.0314
p=0.2745
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

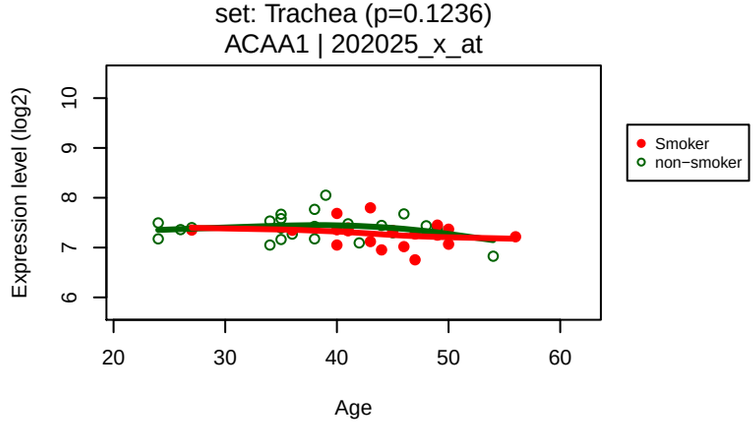

Gene expression ratio in Males
gene: ACAA1 | 202025_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6679
p=0.144
p=0.0383
p=0.5005
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACAA1 | 202025_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4211
p=0.0082
p=0.7849
p=0.4128
SmallAirway2
LargeAirway2
Trachea
Macrophage2

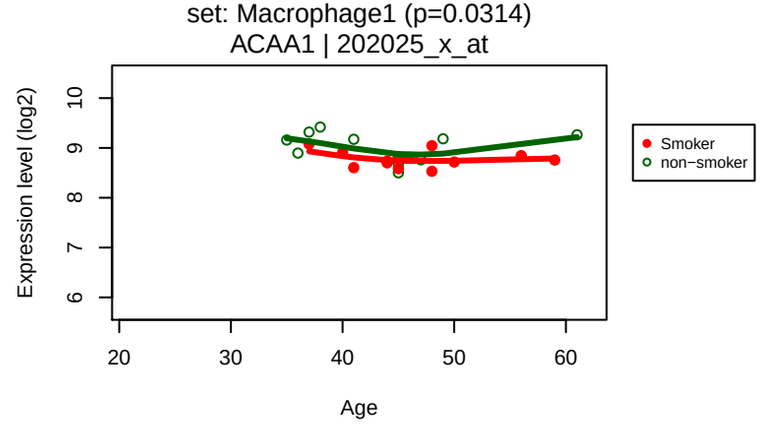

set: SmallAirway1 (p=0.6402)
ACAA1 | 202025_x_at
Expression level (log2)
Age
Smoker
non−smoker

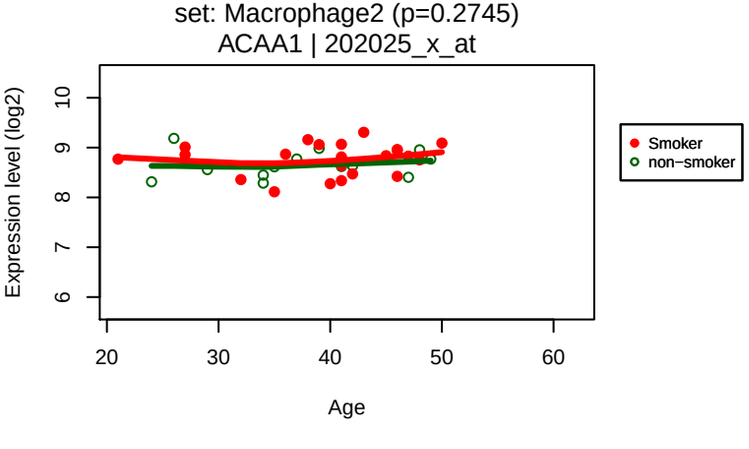

set: SmallAirway2 (p=0.2549)
ACAA1 | 202025_x_at
Expression level (log2)
Age
Smoker
non−smoker

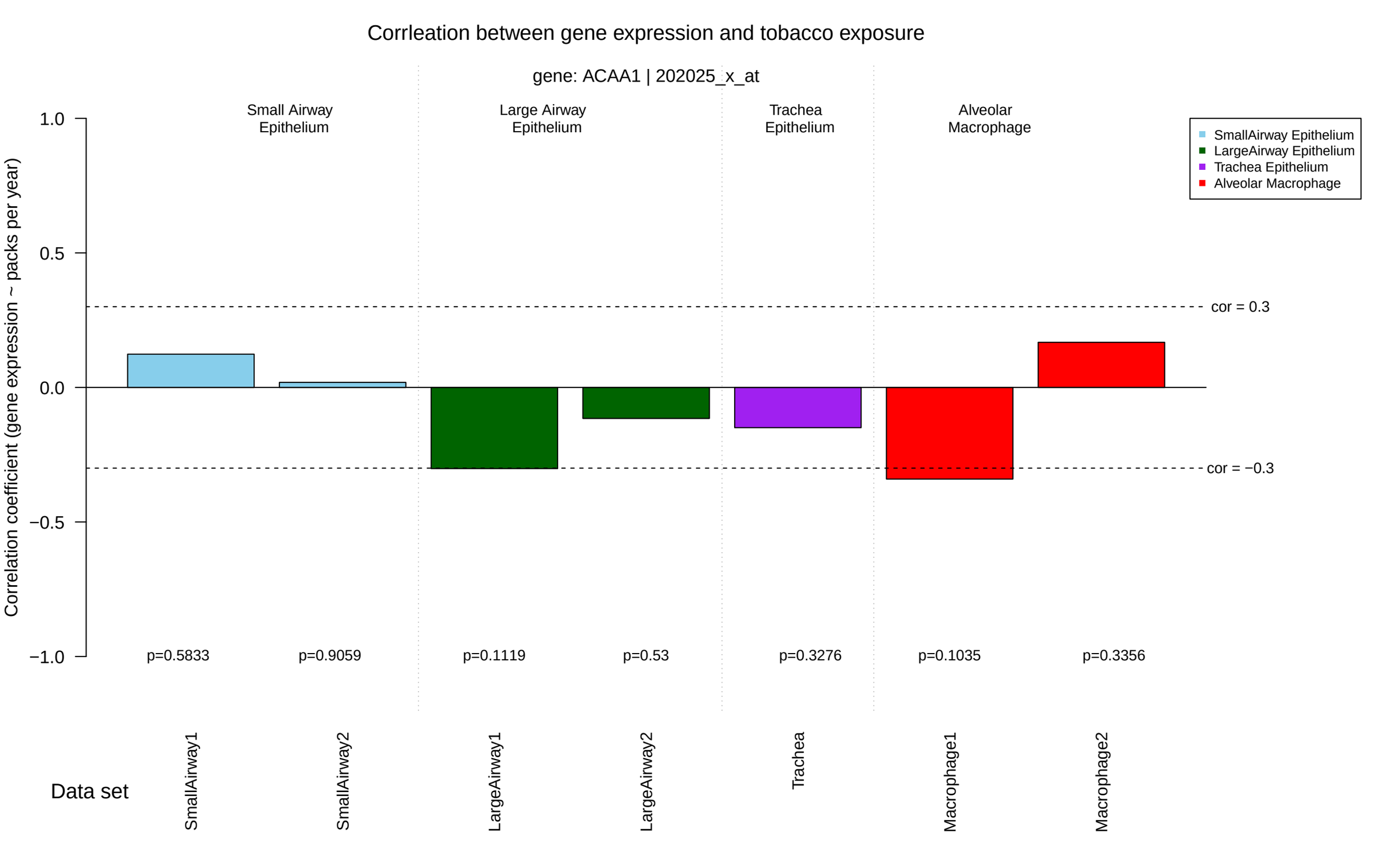

set: LargeAirway1 (p=0.0151)
ACAA1 | 202025_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7777)
ACAA1 | 202025_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1236)
ACAA1 | 202025_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0314)
ACAA1 | 202025_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2745)
ACAA1 | 202025_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACAA1 | 202025_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5833
p=0.9059
p=0.1119
p=0.53
p=0.3276
p=0.1035
p=0.3356
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2